Exp No:2

Upload and Analyze the data set given in csv format and perform data preprocessing and visualization

CODE

```
import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

file_path='C:\sales_data.csv

df = pd.read_csv(file_path)

print(df.head())

print(df.isnull().sum())

df['Sales'].fillna(df['Sales'].mean(), inplace=True)

df.dropna(subset=['Product', 'Quantity', 'Region'], inplace-True)

print(df.describe())

product_summary = df.groupby('Product').agg({'Sales': 'sum','Quantity': 'sum'}).reset_index()

print(product_summary)

plt.figure(figsize=(10, 6))

plt.bar(product_summary['Product'], product_summary['Sales'])

plt.xlabel('Product')

plt.ylabel('Total Sales')

plt.title('Total Sales by Product')

plt.show()

df['Date'] = pd.to_datetime(df['Date'])

sales_over_time = df.groupby('Date').agg({'Sales': 'sum'}).reset_index()

plt.figure(figsize=(10, 6))

plt.plot(sales_over_time['Date'],sales_over_time['Sales'])

plt.xlabel('Date')

plt.ylabel('Total Sales')

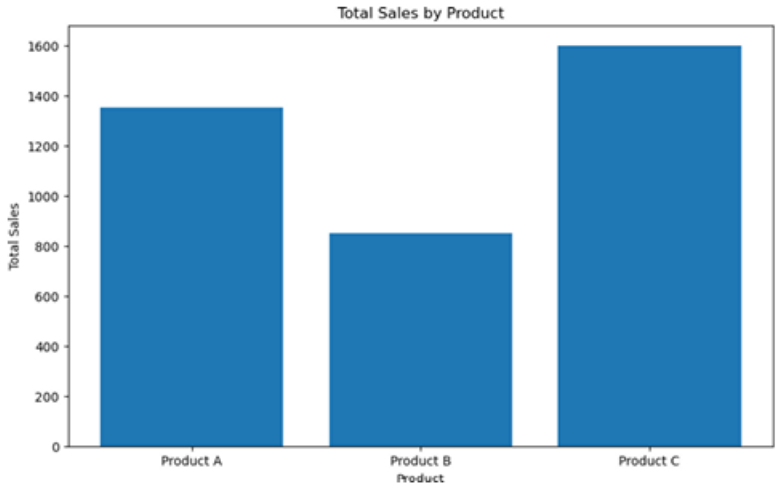
plt.title('SalesOver Time')

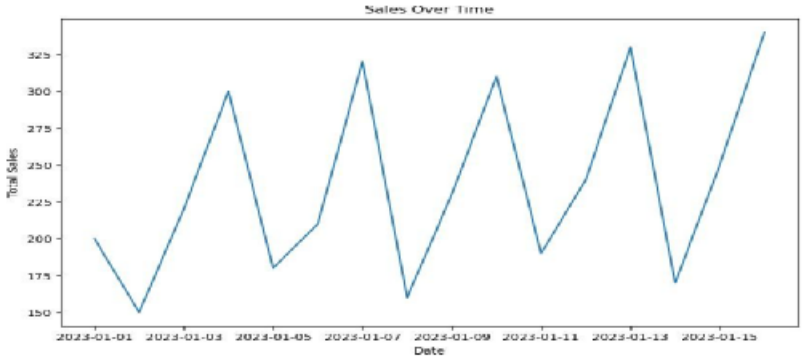
plt.show()
```

```
pivot_table = df.pivot_table(values='Sales', index='Region', columns='Product',
aggfunc=np.sum, fill_value=0)
print(pivot_table)
correlation_matrix = df.corr()
print(correlation_matrix)
import seaborn as sns
plt.figure(figsize=(8, 6))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm')
plt.title('Correlation Matrix')
plt.show()
```

OUTPUT:

```
Date     Product  Sales  Quantity Region
0  2023-01-01  Product A    200         4  North
1  2023-01-02  Product B    150         3  South
2  2023-01-03  Product A    220         5  North
3  2023-01-04  Product C    300         6   East
4  2023-01-05  Product B    180         4   West
Date        0
Product     0
Sales       0
Quantity    0
Region      0
dtype: int64
            Sales   Quantity
count   16.000000  16.000000
mean   237.500000   5.375000
std     64.031242   1.746425
min    150.000000   3.000000
25%    187.500000   4.000000
50%    225.000000   5.500000
75%    302.500000   7.000000
max    340.000000   8.000000
     Product  Sales  Quantity
0  Product A   1350        33
1  Product B    850        17
2  Product C   1600        36
```






```
Product  Product A  Product B  Product C
Region
East             0          0       1600
North         1350          0          0
South            0        480          0
West             0        370          0
              Sales  Quantity
Sales      1.000000  0.944922
Quantity   0.944922  1.000000
C:\Users\Ayyadurai\AppData\Local\Temp\ipykernel_9648\511106317.py:49: Futur
eWarning: The default value of numeric_only in DataFrame.corr is deprecated
. In a future version, it will default to False. Select only valid columns
or specify the value of numeric_only to silence this warning.
  correlation_matrix = df.corr()
```

